VAHE R. JABAGCHOURIAN

ASIC Design Engineer

SUMMARY

Adept technologist and an insatiably inquisitive MSEE degree holder seeking a full-time position as an ASIC hardware design engineer.

ENGINEERING SKILLS

Lab Equipment

Digital Oscilloscope, Frequency Spectrum Analyzer, Digital Multimeter, Logic Analyzer

Hardware Languages

Assembly, C, C++, Verilog HDL

EDA Development / Simulation Tools

Xilinx Vivado, ModelSim , Mentor Graphics ModelSim, Xilinx ISIM, Cadence NCVerilog/NCSim, Xilinx ChipScope

Operating Systems

Microsoft Windows, UNIX, Linux, MacOS

Productivity Software

Word, Excel, PowerPoint

Web Technologies

HTML, PHP, JavaScript, SQL, CSS, ReactJS

RESEARCH PROJECTS

International Digital Terrestrial TV Receiver Design Proposal, and Feasibility Analysis

CSUN Graduate Independent Study

May 2008 Northridge, CA

Researched and proposed a system-level concept for an international Over-The-Air Digital TV receiver using an FPGA in a feasibility document.

Electric Vehicle Battery Equalization

UCI Undergraduate Independent Study

Jan 2003 – Mar 2003 Irvine, CA

Participated in electric vehicle battery management project. Planned, researched equalization, sourced modules, & tested design and validated functionality and performance of vehicle after equalization.

Analog / Digital | Verilog HDL | EDA Tools | Test / Measurement Equipment

Cell: 818 515 7009 Email: vahe.jabagchourian@gmail.com

LinkedIn: https://www.linkedin.com/in/vahe-jab/ Home: West Hills, CA

EDUCATION

M.S. Electrical Engineering — Aug 2007 – Dec 2011

California State University, Northridge — Northridge, CA

Master's Project - Implementation of a 3D Graphics Rendering Pipeline Using an FPGA

https://scholarworks.calstate.edu/downloads/sb397d249

B.S. Computer Engineering — Sep 2001 – Mar 2007

University of California, Irvine — Irvine, CA

Senior Design Project - Voice Activated Presentation Assistant

https://www.scribd.com/document/129977363/Voice-Activated-Presentation-Assistant

UNDERGRADUATE COURSES

Microprocessor Systems
3D Computer Graphics
Digital Systems Design in VHDL
Communication Systems
Compilers & Interpreters
Digital Electronics

GRADUATE COURSES

Verilog HDL for Digital Systems Design
Computer Architecture
3D Scientific Visualization
VLSI Design/Testing
Digital Systems Verification
Computer Arithmetic Design

EXPERIENCE

PHP Developer and IT Assistant — Sep 2020 – Jan 2021

American Business Insurance Services Inc — Westlake Village, CA

- Produced a dynamic sorter in JavaScript for reordering HTML of tabbed report links by name alphabetically reducing report link search time
- Corrected driver ordering scripts in PHP to prevent reorder of driver records 60 days before, resulting in savings of thousands of dollars of extra order cost
- Added ability to highlight excluded drivers across PHP forms, and allowed quick re-query of exclusions on driver changes, using scheduled task
- Created plugged in, unplugged date/time list per vehicle of trackers for vehicles based on vehicle VINs and added time zones to help CSRs (Customer Service Representatives)

Web Applications Developer — **Second Time (#2)**: Jun 2016 – Jan 2018

Rinsoft LLC - Calabasas, CA — **First Time (#1)**: Jun 2014 – Jun 2015

#2A: Created front end-templates for an inventory management system from UI/UX designer using mockups/wireframes within 1 to 2 weeks from inception to completion.

#2B: Implemented C#/ASP.NET/MSSQL Auto Ordering API for automatic order execution, after diagramming PowerPoint flow chart for program logic

#2C: Communicated regular updates of tasks/issues to project manager, developers, stakeholders and finally clients during integration, every day

#2D: Developed REST API integration documentation and high-level computer-aided diagrams for showing API data and logic flow, using draw.io and MS PowerPoint

#2E: Accomplished integration with LightSpeed picking software and VendSys ordering API in 6 months, demoing results to vendor/client

#1A: Reduced graphic artists' project cycle time via developed CSS/AngularJS/Masonry.js dashboard for optimizing graphic artist project workflow

#1B: Made a 10-worksheet pivot table for 10 round 2016 FCC reverse incentive auction in Excel from one table https://www.fcc.gov/about-fcc/fcc-initiatives/incentive-auctions/how-it-works

#1C: Fox Networks Engineering and Operations directly utilized produced deliverables across 3 different departments - scheduling, engineering and graphics

PERSONAL QUALIFICATIONS

- Architecturally minded, systematic, logical, and habitual in planning, performing, and verifying accuracy of work
- Quickly able to learn new subjects
- Eager and excited to receive corrective feedback for immediate improvement
- Continuously self-aware and self-initiative
- Purposefully driven to accomplish multiple goals
- Meticulous, persistent, and relentlessly attentive to small details
- Endlessly enthusiastic and energetic
- Excellent with numbers
- Strongly driven by a sense of time urgency in the matters of immediate priority demanding timely completion
- Charismatic & naturally able to relate to a wide variety of individuals at their own level
- Strongly, openly, communicative and articulate in technical, non-technical and empathetic matters
- Naturally inclined and hungry to lead by example, and by service
- Persistent, persuasive and able to bring out the very best in others
- Methodically & strategically organized
- Insatiably curious and inquisitive about computers, electronics, and technology
- Capable of working both independently and within team environments

GRADUATE VLSI PROJECT

Analog To Digital Converter
University of California, Irvine
Aug 2005 – Mar 2006 Irvine, CA

Designed and simulated, tested and validated physical realization of a 6-bit Flash Analog to Digital Converter (ADC) using Magic, IRSIM and HSPICE, and electronic lab measurement equipment.

EXPERIENCE

Software Engineering Intern Nov 2009 – Sep 2010
ITT Incorporated - Radar Systems: Gilfillan Van Nuys, CA

- Automated generation of embedded tables/images in a dynamic Interface Control Document (ICD) generator using PHP, MySQL, and phpRtfLite
- Developed a PHP/MySQL (MyISAM Engine) History Log viewer on multi-million records, paginating result for quick load (seconds), with savable queries
- Reduced time for resizing images to fit within page margins of RTF (Rich Text Format) documents resulting in reduction of automating manual resize time from hours to 30 minutes
- Created PHP scripts to search Radar cable from start/end points in PHP/MySQL, enabling quick search of cable locations from PDF cable trace files
- Manually loaded all textual documentation for Maintenance Personal Computer (MPC) desktop application features into MySQL in 30 minutes
- ITT used produced deliverables for a live demonstration to customer for a coastal radar upgrade project during an on-site factory acceptance test

HARDWARE DEVELOPMENT PROJECTS

VAPA [Voice Activated Presentation Assistant] Aug 2006 - Mar 2007
University of California, Irvine Irvine, CA
https://www.yumpu.com/en/document/view/30414803/voice-activated-presentation-assistant-eecs-senior-design-project

- Designed and constructed a Voice Activated PowerPoint Controller incorporating the HM2007 speech recognition board, VersaPoint transmitter, Arduino microcontroller, and countdown display.
- Created a driver program in the C programming language to enable the system's functionality.
- Utilized schematic capture software (EaglePCB) to generate technical computer-aided drawings, illustrating the complete system at both the chip and component level, and these drawings were exported for inclusion in the final project report.
- The final project report served as a high-quality model and was employed by the UCI EECS department to directly support ABET (Accreditation Board for Engineering Technology) accreditation

Xilinx Virtex-7 VC707 Based FPGA GPU Implementation Jan 2023 – Present
https://github.com/vahejab/arty_sv_sampler West Hills, CA
Documented Q/A (Jan 2023 – March 2023) available on Electronics Stack Exchange at https://electronics.stackexchange.com/users/20332/vahe?tab=questions

- Completed development of PS/2 mouse input module for a 3D Graphics Accelerator in 8 weeks using FPro (FPGA Prototype) template design from Dr. Pong Chu's prototype example in FPGA Prototyping by SystemVerilog examples (Xilinx MicroBlaze MCS SoC Edition) textbook
- Performed behavioral and timing simulation on ModelSim to identify primary FIFO issues and resolved them for both UART and PS/2 mouse modules
- Conducted synthesis and validation on Xilinx VC707 FPGA Evaluation board while connected to PS/2 mouse and USB UART with host PC
- Used PuTTY terminal to observe real-time mouse movement packets from a 3D trackball across three axes and monitored button presses
- Located an appropriate Level I2C (Inter Integrated Circuit) shifter for bidirectional voltage shifting between +1.8 Volts and +5 Volts
- Assembled the Level Shifter on a Digilent PMOD BB (Breadboard) and connected it to a Digilent, Inc. FMC-CE expansion board, which plugs onto the Virtex-7 evaluation board via one of two FMC Mezzanine connectors

VOLUNTEER WORK

Equipment Re-Racker Nov 2021 – Present

Crunch Gym Chatsworth, CA

- Conducted individual daily, weekly and increasingly frequent and regular equipment re-rack and organization efforts for the direct benefit of the entire facility (management, employees, and members).
- Expanded voluntary efforts two other sites (Northridge, CA, and newly acquired West Hills, CA facility - formerly Athletic Society.
- Achieved complete re-rack organization and site cleanup for full facility on average of 5 hours on each visit.
- Organization efforts resulted in significant and noticeable improvements in facility member reviews/ratings reaching 4/5 stars.

EXTRA PROJECTS

Spacecraft Testbed Power Loss Calculator

Jan 2015 - Feb 2015 West Hills, CA

- Created a web application capable of allowing engineers to quickly see and calculate power loss in decibels(dB) across any two nodes from cables on electronic assemblies of a spacecraft.
- Performed application development in HTML, and JavaScript along and produced three local files to providing power levels for parsing.

FIND ME ONLINE

- Electronics StackExchange Q&A Profile

https://stackoverflow.com/users/1691103/vahe

- GitHub

https://github.com/vahejab

- LinkedIn

https://www.linkedin.com/in/vahe-jab/

WEB APPLICATION DEVELOPMENT PROJECTS

Risk Management Web Application 2022 – Present

https://github.com/vahejab/RiskAIM-Overview

- Produced a single Angular.JS project risk management web application to track and reduce project risks via REST API-centric architecture
- Updated controllers to a component-based framework in ReactJS and upgraded RESTFUL API to a feature-based vertical slice architecture / application directory structure
- Achieved creation of an advanced adaptive tracking form to track risk mitigation events using waterfall metrics, and PowerPoint reports with a 5 by 5 risk matrix showing risk level mitigation progress
- Completed development of a data visualization dashboard grid using Gridster.js (Front End JavaScript Grid Library), Angular-Gridster.js, d3.js (Data Driven Documents), and dc.js (Dimensional Charting) library
- Utilized HTML5 for dropdowns, date picker for calendar control, and pre-existing sample tabular code for producing responsive tables
- Accomplished construction of visualization mockup for dashboard data visualization of entire project risks (specifically buildup, burndown, and bar-chart metrics)
- Incorporated PHP (Hypertext Preprocessor) Composer library for dependency management as well as latest features of PHP8 for annotation-based REST (Representational State Transfer) URL (Uniform Resource Locator) routing
- Leveraged PHP OpenTBS (Tiny But Strong) for report generation, and PHPOffice for generation of PowerPoint graphs, 5 by 5 risk matrix, and a risk mitigation waterfall chart

Action Item Management Web Application 2013 – 2017

https://www.ieee-bv.org/wp-content/uploads/2016/07/newsletter_2016_07.html#b4

- Created an action item management app in PHP/MySQL along with D3.js dashboard status charts (build-up, burn-down, stacked bar)
- Incorporated sortable summary page with red colored indicators on dates for past due action items and allowed item searching/filtering
- Programmed functionality to activate, schedule, and send automatic email reminders of past due and approaching action items
- Delivered a presentation/demonstration for IEEE (Institute of Electrical and Electronics Engineers) Buenaventura Section
- Demonstrated complete functionality of application which was loaded onto an AWS (Amazon Web Service) hosting environment configured for public viewing

LANGUAGE PROFICIENCY

English	Native (Speaker, Reader, Writer, Understander)
Armenian	Native (Speaker, Reader, Writer, Understander)
Spanish	Advanced (Speaker, Reader, Understander)